

LUIS ROMERO

AI Engineer | Backend Developer (Node.js/TypeScript)

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PROFESSIONAL SUMMARY

Backend Developer / AI Engineer with 3 years of experience building and operating AI products in production. I founded and operate Cronix, a 24/7 multi-tenant SaaS with 150+ real appointments managed autonomously by a custom, no-framework AI orchestration engine for WhatsApp and voice (restricted tool calling), backed by a deterministic anti-hallucination architecture and an end-to-end in-house evaluation (DeepEval) and observability (LangSmith) system. I also built the AI backend of an institutional platform with a conversational assistant, RAG and semantic search over pgvector (LangGraph, IBIME), and of an industrial CMMS; I integrated payments, OAuth, real-time chat and video in a teletherapy platform. Core stack: Node.js, TypeScript, PostgreSQL/SQL, serverless and event-driven architecture.

EXPERIENCE

Backend / Full-Stack Developer | IBIME (Institutional platform, Remote)

January 2025 - Present

- Designed, built and deployed end-to-end the digital platform for a public library network (replacing a legacy HTML/CSS system), with an AI assistant for citizen queries; adopted by the institution, with requirements gathered with librarians and the institution's IT team.
- Designed a hybrid chat engine with a 5-layer anti-hallucination architecture (Intent Classifier, Privacy Gate, fail-hard RAG threshold, Response Guardrail and Response Policy) that keeps the LLM out of the critical flow: when the email is known it answers directly from the database, removing LLM calls and hallucinations on those queries.
- Implemented RAG and semantic search over PostgreSQL + pgvector (Gemini embeddings) with Redis caching and a custom rate limiter with graceful degradation, plus a stateful LangGraph agent (extractor-validator-corrector) that structures catalog PDFs into validated JSON with fail-closed duplicate checks.
- Applied hexagonal architecture and dependency injection with SOLID principles and shared front-back Zod; CI with 230+ tests (Vitest + Playwright); deployed on Render + Vercel.
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Backend Developer | Kiura (Barranquilla, Colombia, Remote)

April 2024 - December 2024

- Built backend services for a teletherapy platform (patients-therapists) unifying subscription payments, appointments, real-time messaging and video sessions; deployed on Render with a relational database provisioned on AWS (RDS).
- Integrated MercadoPago subscription payments (creation, cancellation, webhook) that activate or suspend access, real-time chat with Socket.IO and video sessions with signed, per-room Agora RTC tokens.
- Implemented dual authentication (JWT + Google OAuth) with RBAC over a 26-entity relational domain in MySQL/Sequelize with scheduling and conflict detection.

PROJECTS

Founder & AI / Backend Engineer | Cronix

January 2026 – Present

Multi-tenant SaaS for appointment-based businesses, built and deployed end-to-end autonomously (DB, serverless backend, AI and infrastructure); in production 24/7 with 150+ real appointments managed autonomously and automated reminders via pg_cron, no manual intervention.

- Built the conversational agent that books, reschedules and cancels appointments over WhatsApp and voice, autonomously and 24/7 the client handles it simply by speaking, no human intervention required (custom orchestration engine, restricted tool calling, Groq + Deepgram, 12 capabilities).
- Designed the controls that stop the agent from hallucinating or acting on the wrong client or appointment — the main risk of connecting an LLM to a real database: a deterministic 6-layer / ~13-mechanism architecture (fast paths, 2-turn confirmation gate and function calling restricted to declared args, among others).
- Built the system that measures and monitors the agent's reliability: a regression suite with LLM-as-judge (DeepEval) over versioned golden datasets, integrated into CI, and multi-tenant tracing (PostgreSQL + export to LangSmith) with a p50/p95 latency dashboard per business, across 9 serverless Edge Functions (Deno).
- Isolated each business's data with PostgreSQL RLS (59 policies / 32 tables, validated by 138 pgTAP assertions) and hybrid JWT + WebAuthn/Passkeys auth with rate limiting on Redis; integrated 4 payment methods (PayPal, NOWPayments, Pago Movil, Binance) with idempotent fulfillment and async processing via QStash.
- Maintained quality with a 1,000+ test suite (Vitest, Playwright E2E, pgTAP), CI/CD (GitHub Actions + Vercel) with Sentry and Spec-Driven Development with CI drift checks; integrated MCP servers (PayPal, Supabase, LangSmith) as agent tooling.

CMMS Hidrobombas Merida (Remote) | Full-Stack Developer

April 2026 - Present

- Designed and shipped end-to-end a CMMS (52 REST endpoints over 11 data models) that replaced an industrial maintenance company's paper inspection sheet; requirements gathered with the client in iterative deliveries.
- Built an AI assistant with multi-agent orchestration (LangGraph): a tool-calling agent + a multi-node diagnostic graph, with RAG over the report history; AI layer testable via dependency injection and a pluggable vector store (in-memory / pgvector).
- Implemented JWT auth with rotating refresh and server-side revocation, role + ownership authorization, idempotency, rate limiting and SSE streaming; also delivered a React 19 PWA with offline mode (IndexedDB + Background Sync) and PDF reports; 394 tests (82% backend line coverage) and CI with dual DB support (PostgreSQL/Neon + SQLite).

Twitter/X Clone (technical challenge) | Full-Stack Developer 2026

- Implemented hardened custom auth (pinned-HS256 JWT, per-request DB user validation, bcrypt and a production boot guard against weak secrets) and real-time over Server-Sent Events with topic-based pub/sub, with no polling.
- Modeled the social graph with composite DB-level uniqueness (idempotent follow/like) over Prisma with SQLite (dev) and PostgreSQL (Docker); a multi-service Docker Compose stack and a 3-tier test pyramid (138 integration Supertest, 54 components, 10 E2E Playwright).

TECHNICAL SKILLS

Backend: Node.js, TypeScript, Express.js, REST API, RESTful APIs, Serverless, Edge Functions, Deno, Event-Driven Architecture, Webhooks, Webhook Processing, Idempotency, SOLID Principles, OOP, Dependency Injection, Hexagonal Architecture, Socket.IO, WebSockets, Agora RTC

AI: LLM Integration, LangChain, LangGraph, Agentic Systems, Multi-Agent Orchestration, Tool Calling, Function Calling, Retrieval-Augmented Generation (RAG), Semantic Search, Vector Databases (pgvector), Anti-Hallucination, LLM Evaluation (DeepEval), LLM Observability (LangSmith), Model Context Protocol (MCP), Voice AI, Groq, Gemini, Hugging Face, Deepgram, OpenAI-compatible API, n8n, Workflow Automation

Databases: PostgreSQL, SQL, AWS (RDS), MySQL, Sequelize, Supabase, SQLite, Redis, Upstash Redis, pgvector, Row-Level Security, Multi-tenant

Security/Auth: JWT, OAuth 2.0, WebAuthn, Passkeys, RBAC, Rate Limiting

Payments/Async: PayPal REST API, MercadoPago, Payment Integration, Upstash QStash, Async Processing, Message Queues, Server-Sent Events (SSE), Circuit Breaker

DevOps/Testing: Docker, GitHub Actions, CI/CD, Spec-Driven Development, Vercel, Render, Sentry, Observability, Vitest, Jest, Unit Testing, Integration Testing, E2E Testing, Playwright, pgTAP, fast-check

Frontend: React, Next.js 15, App Router, Server Actions, SSR, Tailwind CSS, TanStack Query, Zod, PWA, Service Workers, i18n

EDUCATION

- Bachelor's Degree in Systems Engineering (Information Systems)
Universidad Politécnica Territorial de Mérida Kléber Ramírez (UPTM) - Expected Graduation: Dec 2026
- Full-Stack Bootcamp | Academlo - Mar 2024 (Node.js, React, PostgreSQL, Express, HTML, CSS)

LANGUAGES

Spanish: Native

English: Intermediate (B1+) | Advanced technical reading/writing; professional communication improving